

VINICIUS MACHIONI

Front-End Engineer | Front-End Developer | React, Next.js, TypeScript, Node.js
Poços de Caldas, MG, Brazil | Open to 100% remote roles worldwide | Employee, PJ or contractor
machioni.dev@gmail.com | +55 (35) 99810-4311
linkedin.com/in/viniciusmachioni | github.com/viniciusmachioni | machioni.dev.br

PROFESSIONAL SUMMARY

Front-end developer building production web applications since 2023 with React, Next.js and TypeScript, and working as a full-stack developer when a feature needs an API, a database, real-time updates or a deploy pipeline. Currently building the React front end of IAtrix, a SaaS management platform, against a C# and .NET back end. Previously led the redesign of the PUC Minas institutional portal, live today at pucpcaldas.br. Specialized in web accessibility (WCAG 2.1) and Core Web Vitals performance. B.Sc. in Computer Science, with postgraduate studies in Software Architecture in progress. Native Portuguese and advanced English.

TECHNICAL SKILLS

Front-End: React, Next.js, App Router, TypeScript, JavaScript, ES6+, HTML5, CSS3, Vue.js, Zustand, Vite, Tailwind CSS, Radix UI, TanStack Router, TanStack Query, React Hook Form, Zod, single-page applications, responsive design, component libraries, state management

Accessibility and Performance: WCAG 2.1, axe-core, ARIA, keyboard navigation, screen reader compatibility, color contrast auditing, Core Web Vitals, LCP, CLS, lazy loading, code splitting, rendering strategy

Back-End: Node.js, Express, Fastify, REST APIs, WebSockets, Socket.io, Prisma, pg-boss, C#, .NET, Laravel, PHP, Ruby on Rails

Databases: PostgreSQL, MySQL, SQL, query optimization, index optimization

Testing and Quality: Playwright, Vitest, React Testing Library, end-to-end testing, code review, Clean Code, SOLID

Cloud and DevOps: Microsoft Azure, AWS, Google Cloud Platform, Docker, Docker Compose, Git, GitHub Actions, GitLab CI/CD, Bitbucket, Turborepo, pnpm, monorepo

Integrations and AI: Stripe, Google Gemini API, Puppeteer, WhatsApp Business API, Instagram API, OAuth 2.0, Firebase, Model Context Protocol (MCP)

PROFESSIONAL EXPERIENCE

Full-Stack Developer

GrandPneus, Poços de Caldas, Brazil

July 2026 - Present

- Build the React and TypeScript front end of IAtrix (iatrix.com.br), a SaaS management platform for auto repair shops, covering AI photo-based service validation, invoicing, financial and inventory modules, image upload, and audio recording and playback.
- Built a customer service CRM interface from scratch in React, including WhatsApp and Instagram integrations, social post and ad creation, and role-based access control.
- Integrate the React front end with a C# and .NET back end over REST APIs, delivering features end to end.
- Work across Microsoft Azure, AWS and Google Cloud Platform in the deployment and integration of the platform's services.
- Gather requirements directly with business stakeholders and prioritize the backlog around the operational routine of the shops.
- Technologies: React, TypeScript, REST APIs, C#, .NET, Microsoft Azure, AWS, Google Cloud Platform

Full-Stack Developer

PUC Minas, Poços de Caldas, Brazil

July 2025 - June 2026

- Led the redesign of the institutional portal (pucpcaldas.br) in React and TypeScript, live today and used by students, faculty and applicants.
- Reduced Largest Contentful Paint (LCP) on the main pages to under 2 seconds through lazy loading and a revised rendering strategy.
- Built the interface with Tailwind CSS and a reusable component structure later adopted across the team.
- Introduced Clean Code, SOLID and WCAG 2.1 accessibility criteria into the team code review process.
- Implemented CI/CD pipelines in GitLab, shortening the deploy cycle and reducing manual release errors.
- Technologies: React, TypeScript, Tailwind CSS, Laravel, MySQL, Docker, GitLab CI/CD

Full-Stack Developer, Internship

Donato Hospital de Olhos, Poços de Caldas, Brazil

January 2023 - June 2024

- Built clinical dashboards in Vue.js with lazy loading and Vuex state management, cutting render time on high-volume screens.
- Refactored PostgreSQL indexes to keep critical clinical dashboards responsive under load.
- Contributed to digitizing clinical processes previously handled on paper, shortening average patient turnaround time.
- Standardized development and production environments with Docker and Docker Compose, reducing deploy errors.
- Technologies: Vue.js, Vuex, Ruby on Rails, PostgreSQL, Docker

KEY PROJECTS

Gabarita, question bank for medical residency exams, in production (gabaritamed.com)

- Nearly 18,000 questions ingested from official exam PDFs and organized into a medical taxonomy, running as a paid monthly subscription with Stripe, error monitoring in production and automated backups.
- Spaced-repetition study flow, timed mock exams, review panel with an error log, and AI question generation, with background jobs handled by pg-boss.
- Technologies: Next.js, Prisma, PostgreSQL, pg-boss, Stripe, Google Cloud Platform

Ally-AI, AI-powered web accessibility auditing SaaS (github.com/viniciusmachioni/ally-ai)

- Turborepo and pnpm monorepo with a shared TypeScript types package across front end and back end, plus a Next.js App Router and Tailwind CSS interface with an audit dashboard.
- Audit engine over headless Chrome with Puppeteer and axe-core, detecting contrast, ARIA and keyboard-focus failures against WCAG 2.1, with Google Gemini turning each violation into a concrete code fix.
- Cut token usage per audit by around 60% by batching all violations from a page into a single payload, replacing one LLM round trip per violation.
- Credit-based billing with Stripe webhooks and live updates over Socket.io.
- Technologies: Next.js, TypeScript, Node.js, Express, Puppeteer, axe-core, Gemini API, Stripe, Socket.io, Docker

Flowboard, real-time collaborative board (github.com/viniciusmachioni/flowboard)

- WebSocket synchronization between users editing the same board simultaneously, with Playwright end-to-end tests validating state consistency across two concurrent sessions.
- Technologies: Next.js, Fastify, WebSocket, Playwright, TypeScript

Slotly, booking and scheduling engine (github.com/viniciusmachioni/slotly)

- Server-side availability engine that computes free slots and prevents double booking in real time, converting server UTC to the user local time zone to avoid cross-region conflicts.
- React single-page application with TanStack Router and TanStack Query, Zod validation and accessible Radix UI components.
- Technologies: React, Vite, TypeScript, TanStack Router, TanStack Query, Zod, Radix UI, Laravel, MySQL

Viaja + Chris, CRM for a travel agency, in production (viajamaismcomchris.com.br)

- Lead management, status tracking and a metrics dashboard, built and shipped for a paying client.
- Technologies: Next.js, Prisma, Tailwind CSS

EDUCATION

Postgraduate Certificate, Software Architecture, Data Science and Cybersecurity

PUC-Campinas and PUCPR, dual certification

April 2026 - April 2027, in progress

Bachelor of Science, Computer Science

PUC Minas, Poços de Caldas, Brazil

August 2021 - August 2025

- Thesis: Accessible Web, a Chrome extension for real-time WCAG accessibility auditing with axe-core and AI-generated remediation notes (github.com/viniciusmachioni/accessible-web).

Technical Degree, Information Technology

IFSULDEMINAS, Poços de Caldas, Brazil

January 2018 - December 2020

CERTIFICATIONS AND AWARDS

- Claude Code in Action, Anthropic, March 2026. AI agents and Model Context Protocol (MCP).
- ReactJS and Clean Code Specialization, Rocketseat, February 2026.
- Brazilian Informatics Olympiad (OBI), Brazilian Computer Society (SBC), 2021.
- Third place, Internal Programming Olympiad (OLIP) 2019 and 2020, and IFSULDEMINAS Hackathon 2019, web development track.

LANGUAGES

Portuguese: native. English: advanced, technical communication written and spoken.